

TRAN MINH HUY

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Education

Ho Chi Minh City University of Technology (HCMUT – VNUHCM) Ho Chi Minh City, Vietnam
Bachelor of Computer Science (Japan-Oriented Program) Sep. 2023 – Present

- Course Grades: Calculus 2 (8.5), Machine Learning (9.0), Introduction to Artificial Intelligence (8.6), Software Engineering (8.6) [\[Academic Transcript\]](#)
- **GPA: 3.1/4.0**

Experience

Teaching Assistant – Python & Machine Learning Course Jul. 2025 – Aug. 2025
HCMUT EE Machine Learning & IoT Lab Ho Chi Minh City, Vietnam

- Assisted in teaching Python programming and core Machine Learning concepts to high-school and undergraduate students.
- Supported assignments and student ML projects.
- Mentored student capstone projects, guiding teams through full ML development cycles. [\[Project 1\]](#) [\[Project 2\]](#) [\[Project 3\]](#)

Projects

Multimodal Video Retrieval System Jun. 2025 – Dec. 2025

- Reformulated video search by extracting keyframes using TransNetV2 for efficient querying and reduced computational load on long videos.
- Extracted multimodal features from keyframes: BEiT-3 and CLIP for semantic embeddings, YOLO/Faster R-CNN for object info, and mixed OCR models such as PaddleOCR/EasyOCR/CRAFT for text detection/recognition.
- Built fast retrieval with FAISS (cosine similarity) on semantic embedding and MongoDB Atlas Fuzzy Search for scalable, fuzzy-tolerant metadata matching.
- Refined queries using Gemini LLM to improve semantic relevance and enhance query representation.
- Designed multimodal and temporal querying algorithms for precise video moment localization.
- [Github Repository](#)

Mushroom Images Classification Feb. 2025 – Apr. 2025

- Performed in-depth Exploratory Data Analysis (EDA) to extract valuable statistical insights, class distributions, and visual characteristics from the dataset, guiding targeted data preprocessing, augmentation, and architecture choices.
- Leveraged state-of-the-art deep learning architectures, including ResNet, EfficientNet, and Vision Transformer (ViT), to train robust models for image processing tasks.
- Applied advanced training optimizations such as Early Stopping, Automatic Mixed Precision (AMP), and Learning Rate Schedulers to accelerate convergence and enhance training stability.
- Combined predictions from multiple individual models using Deep Mutual Learning technique, significantly improving system robustness, generalization, and final predictive accuracy.
- Attaining approximately 96% accuracy on the private test set.
- [Github Repository](#)

Adaptive Hybrid Filtering for Robust 3D Motion Correction in Markerless Motion Capture Sep. 2025 – Dec. 2025

- Proposed an adaptive hybrid filtering framework dynamically combining low-pass, Kalman, and 1-Euro filters based on real-time motion state classification (static, slow, fast).
- Integrated global trajectory estimation for physical plausibility enforcement, including ground plane alignment, metric scale recovery, and footskate correction.
- Achieved 30.9% reduction in jitter, minimal temporal lag (0.033s), and real-time performance at approximately 293 FPS on complex martial arts motion data.
- [Github Repository](#)

Publications

Enhanced Multimodal Video Retrieval System: Integrating Query Expansion and Cross-modal Temporal Event Retrieval [\[Arxiv\]](#)

- Van-Thinh Vo, Minh-Khoi Nguyen, Minh-Huy Tran, Anh-Quan Nguyen-Tran, Duy-Tan Nguyen, Khanh-Loi Nguyen, Anh-Minh Phan
- *The 14th International Symposium on Information and Communication Technology – SOICT 2025*

Awards and Achievements

First Prize at Olympic AI HCMC 2025 [\[Certificate\]](#)

Finalist at AI Challenge HCMC 2025 [\[Certificate\]](#)

Technical Skills

Languages: Python, C++, Java

Frameworks: PyTorch, TensorFlow

Libraries: scikit-learn, pandas, NumPy, Matplotlib, seaborn, OpenCV, Ultralytics (YOLO)

Developer Tools: Git, Google Colab, VS Code, Jira

Scholarships

Japan Create International Scholarship [\[Certificate\]](#)

Merit-based Scholarship, Ho Chi Minh University of Technology (VNU–HCM)

OISP Scholarship, Ho Chi Minh University of Technology (VNU–HCM)

Languages

Japanese: JLPT N3 [\[Certificate\]](#)

Additional Information

Lab Manager, HCMUT Machine Learning & IoT Lab Jan. 2026 – Present

Winter School on Quantum Computing Nov. 21–23, 2025

- Organized by the Vietnam Institute for Advanced Study in Mathematics (VIASM) at HCMUT, VNU–HCM.

Online Courses

- CS231n – Deep Learning for Computer Vision (Stanford)
- CS234 – Reinforcement Learning (Stanford)

Soft Skills

- Proficient in reading and analyzing research papers; strong communication, presentation, and teamwork skills; responsible and proactive working attitude.